

Problem C. Hot Potato

Input file: stdin
Output file: stdout
Time limit: 1 second
Memory limit: 128 MB

In the United States, there is a well-known children's game called **Hot Potato**, where players sit in a circle and rapidly pass an object from one person to the next.

Now, there are n players, numbered from 0 to $n - 1$, sitting in a circle clockwise.

At time $t = 0$, the Hot Potato is initially held by player s .

Whenever a player at position i receives the Hot Potato at time t , we compute

$$r = t \bmod 3$$

According to the value of r , the next player and the waiting time are determined as follows:

- If $r = 0$, the Hot Potato is thrown to the player on the **opposite side of the circle**, i.e. $(i + n/2) \bmod n$.

The player waits $O[i]$ time units before throwing, and the throw itself takes $n/2$ time units.

- If $r = 1$, the Hot Potato is thrown to the **right (clockwise)**, i.e. $(i + 1) \bmod n$.

The player waits $R[i]$ time units before throwing, and the throw takes 1 time unit.

- If $r = 2$, the Hot Potato is thrown to the **left (counter-clockwise)**, i.e. $(i - 1 + n) \bmod n$.

The player waits $L[i]$ time units before throwing, and the throw takes 1 time unit.

If the Hot Potato is received at time t , then the next player receives it at time

$$t' = t + \text{wait}(i, r) + \text{throw}(r).$$

This process continues indefinitely.

Given a time limit T , your task is to determine which player is holding the Hot Potato at time T .

If the Hot Potato arrives at a player exactly at time T , output that player.

If the Hot Potato would be received by the next player strictly after time T , then it remains with the current player.

Input

The input consists of four lines:

- Line 1 contains three integers n, s, T where n is even, $0 \leq s < n \leq 10000$, and $1 \leq T \leq 10^{18}$.
- Line 2 contains n integers $L[0], L[1], \dots, L[n-1]$.
- Line 3 contains n integers $R[0], R[1], \dots, R[n-1]$.
- Line 4 contains n integers $O[0], O[1], \dots, O[n-1]$.

All values in L, R, O satisfy $0 \leq L[i], R[i], O[i] \leq 10^9$.

Output

Print a single integer — the index of the player holding the Hot Potato at time T .

Example 1

stdin	stdout
6 2 10 1 2 1 2 1 2 1 1 2 1 2 1 1 2 1 2 1 2	3

Explanation 1

Initially, player 2 holds the Hot Potato at time $t = 0$.

We simulate the process step by step.

- At time $t = 0$, we have $t \bmod 3 = 0$.

So player 2 throws the Hot Potato to the opposite player:

$$(2 + 6/2) \bmod 6 = 5.$$

Player 2 waits for $O[2] = 1$ time unit, and the throw takes $6/2 = 3$ time units. Therefore, player 5 receives the Hot Potato at

$$0 + 1 + 3 = 4.$$

- At time $t = 4$, we have $t \bmod 3 = 1$.

So player 5 throws the Hot Potato to the right player:

$$(5 + 1) \bmod 6 = 0.$$

Player 5 waits for $R[5] = 1$ time unit, and the throw takes 1 time unit. Therefore, player 0 receives the Hot Potato at

$$4 + 1 + 1 = 6.$$

- At time $t = 6$, we have $t \bmod 3 = 0$.

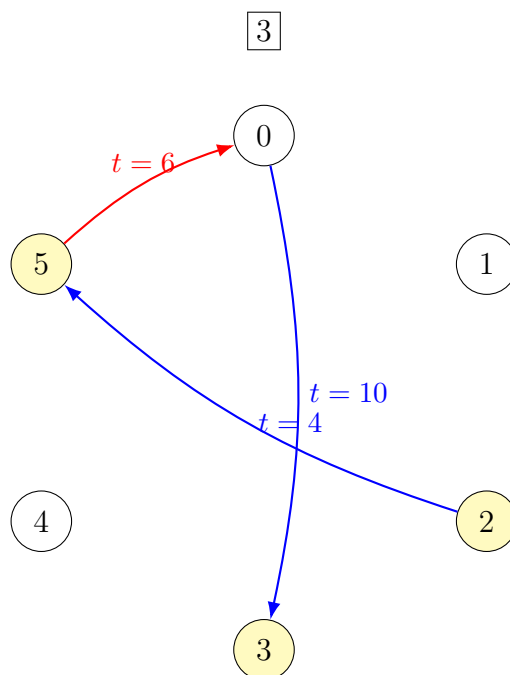
So player 0 throws the Hot Potato to the opposite player:

$$(0 + 6/2) \bmod 6 = 3.$$

Player 0 waits for $O[0] = 1$ time unit, and the throw takes 3 time units. Therefore, player 3 receives the Hot Potato at

$$6 + 1 + 3 = 10.$$

Since the Hot Potato arrives at player 3 exactly at time $T = 10$, the answer is



Blue arrows: opposite move ($t \bmod 3 = 0$)
 Red arrow: right move ($t \bmod 3 = 1$)

Figure 1: Example 1: the Hot Potato moves from $2 \rightarrow 5 \rightarrow 0 \rightarrow 3$

Example 2

stdin	stdout
12 3 100 2 1 3 2 4 2 5 3 1 2 4 1 1 3 2 4 2 5 1 4 2 3 3 2 6 5 4 6 4 6 5 7 6 5 4 6	10

Explanation 2

Whenever a player receives the Hot Potato at time t , the next move is determined by $t \bmod 3$.

Starting from player 3, the Hot Potato eventually reaches player 10. The next throw would make another player receive it after time 100, so at time $T = 100$, the Hot Potato is still held by player 10.